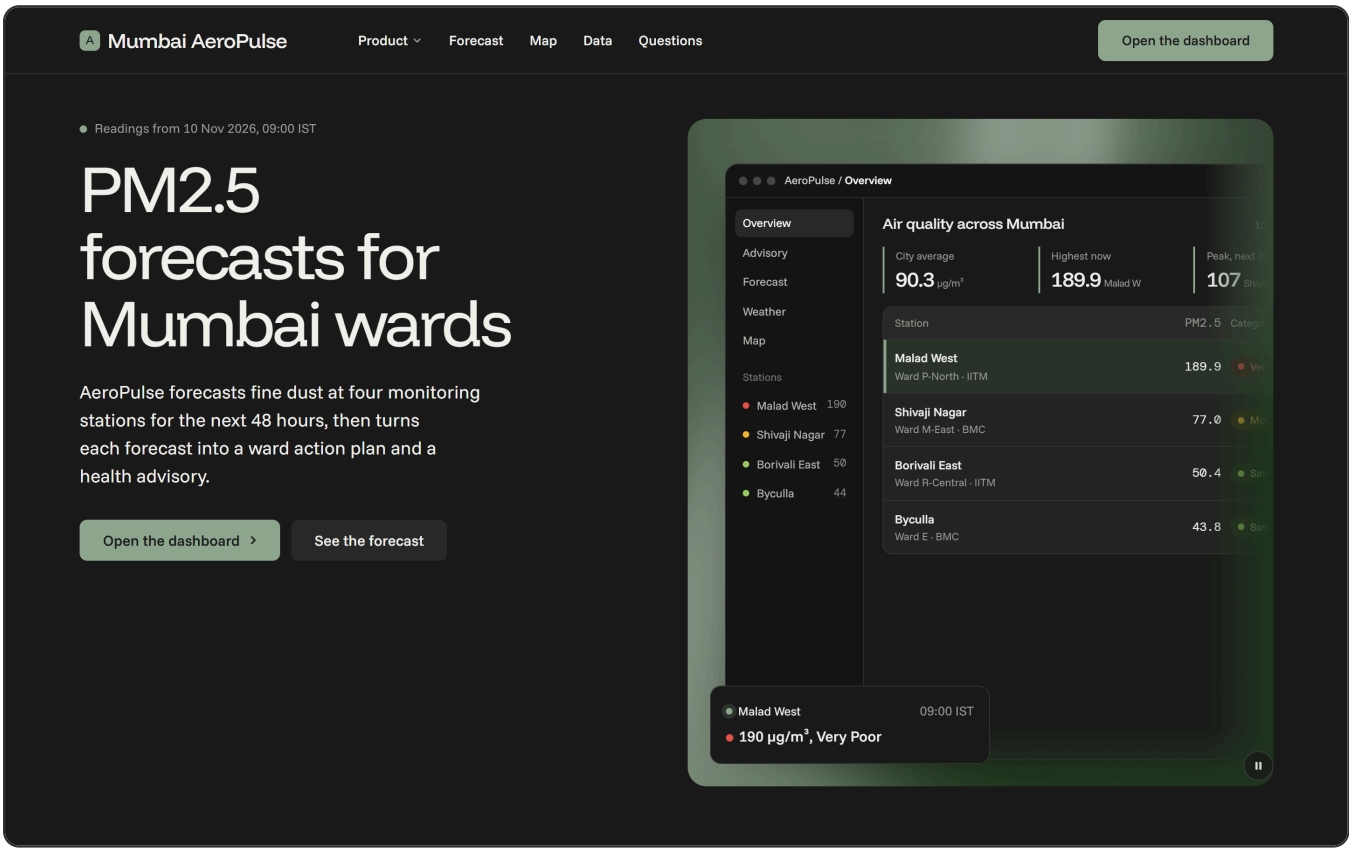
GOOGLE MUMBAI AIR QUALITY MAPPING & FORECASTING CHALLENGE · ZINDI · CREATIVE TRACK

Mumbai AeroPulse

PM2.5 forecasts for four Mumbai monitoring stations, turned into a ward-level action plan for the BMC and a health advisory for residents.

LIVE WEBSITE
https://amanpundhir.github.io/mumbai-aeropulse/

SOURCE CODE
https://github.com/amanpundhir/mumbai-aeropulse



The idea

A forecast on its own does not tell a ward officer where to send a mist cannon, or tell a parent whether the morning walk to school is safe. AeroPulse takes the two models submitted to the challenge, puts them next to what each station measured, adds the weather that explains them, and ends in two plain lists of what to do: one for the municipal ward office, one for the people who live there.

It covers the four stations in the challenge dataset: Borivali East and Malad West (run by IITM) and Byculla and Shivaji Nagar (run by the BMC), each tied to its ward: R-Central, P-North, E and M-East.

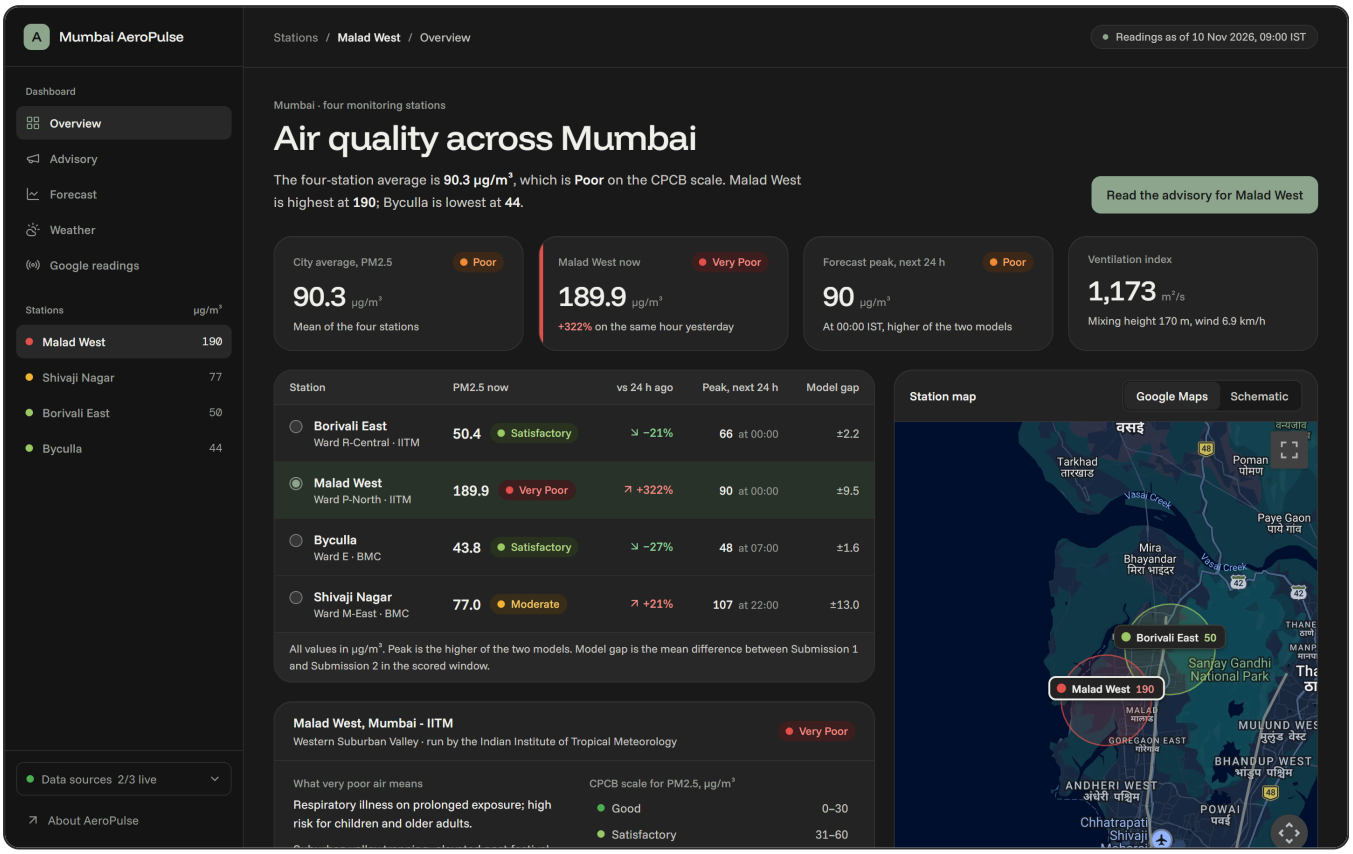
What the site does

Part	What it shows

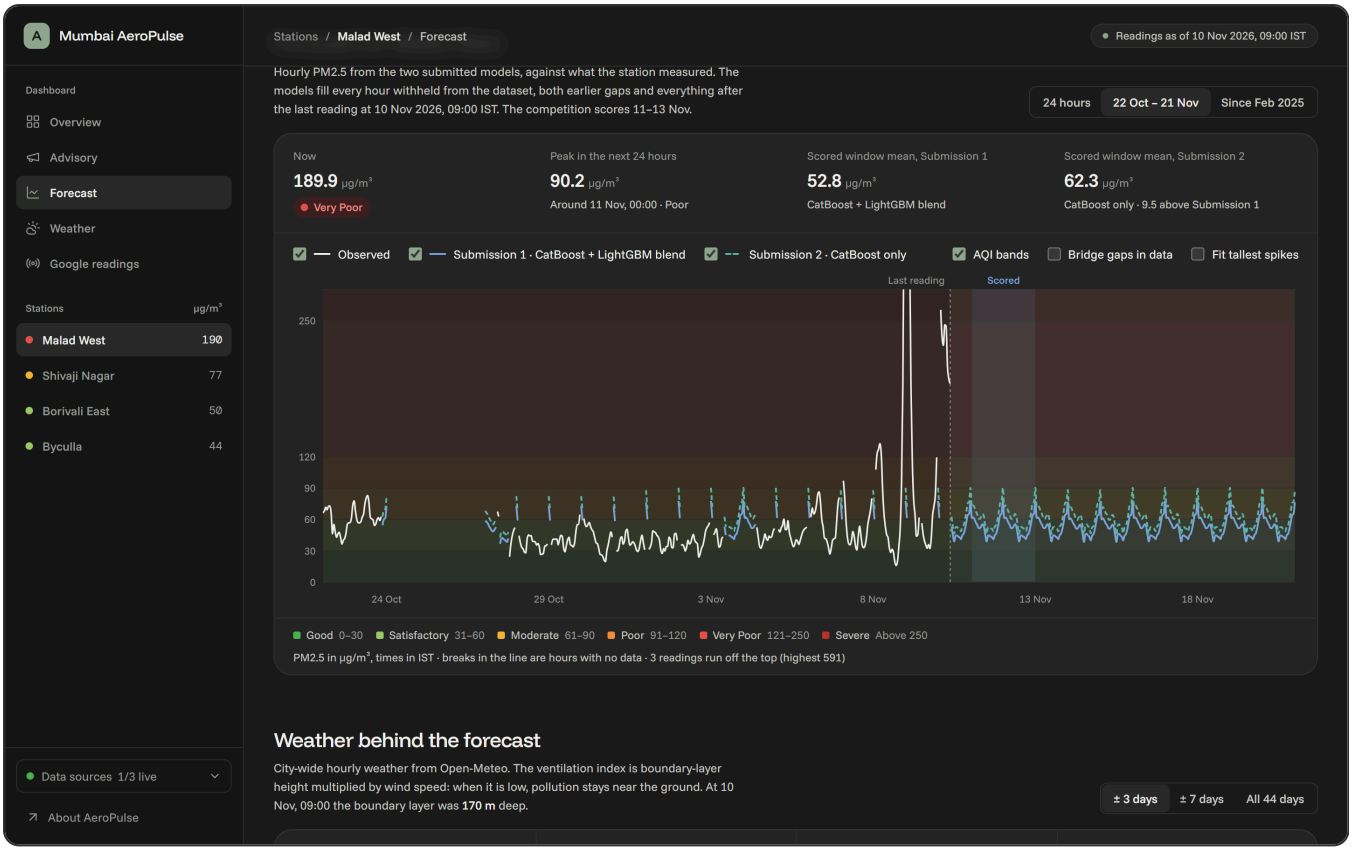
Overview	The city average, the selected station now and against the same hour yesterday, the higher of the two models' peaks for the next 24 hours, and the ventilation index. Every value is placed on the CPCB NAQI PM2.5 scale (Good to Severe).
Station map	All four stations coloured by category. Google Maps when an API key is set; otherwise a built-in map of the island drawn in the site itself.
Forecast	Hourly PM2.5 measured, Submission 1 and Submission 2 on one chart with the CPCB bands, over 24 hours, 22 Oct to 21 Nov, or since February 2025. The scored window (11 to 13 Nov) is marked.
Weather	Open-Meteo boundary-layer height, wind, temperature and humidity, with the ventilation index (boundary-layer height × wind speed), and how each moves with PM2.5 at the station.
Advisory	A municipal action plan with timed, place-specific tasks (for example clean-construction checks at Mindspace for Ward P-North) and advice for residents split by group: everyone, children, older adults, people with asthma.
Google readings	Google Air Quality API values for the station's location as a cross-check on the ground sensor, with Google's health guidance.

Screens

Dashboard overview



Forecast: measured PM2.5 against both submitted models

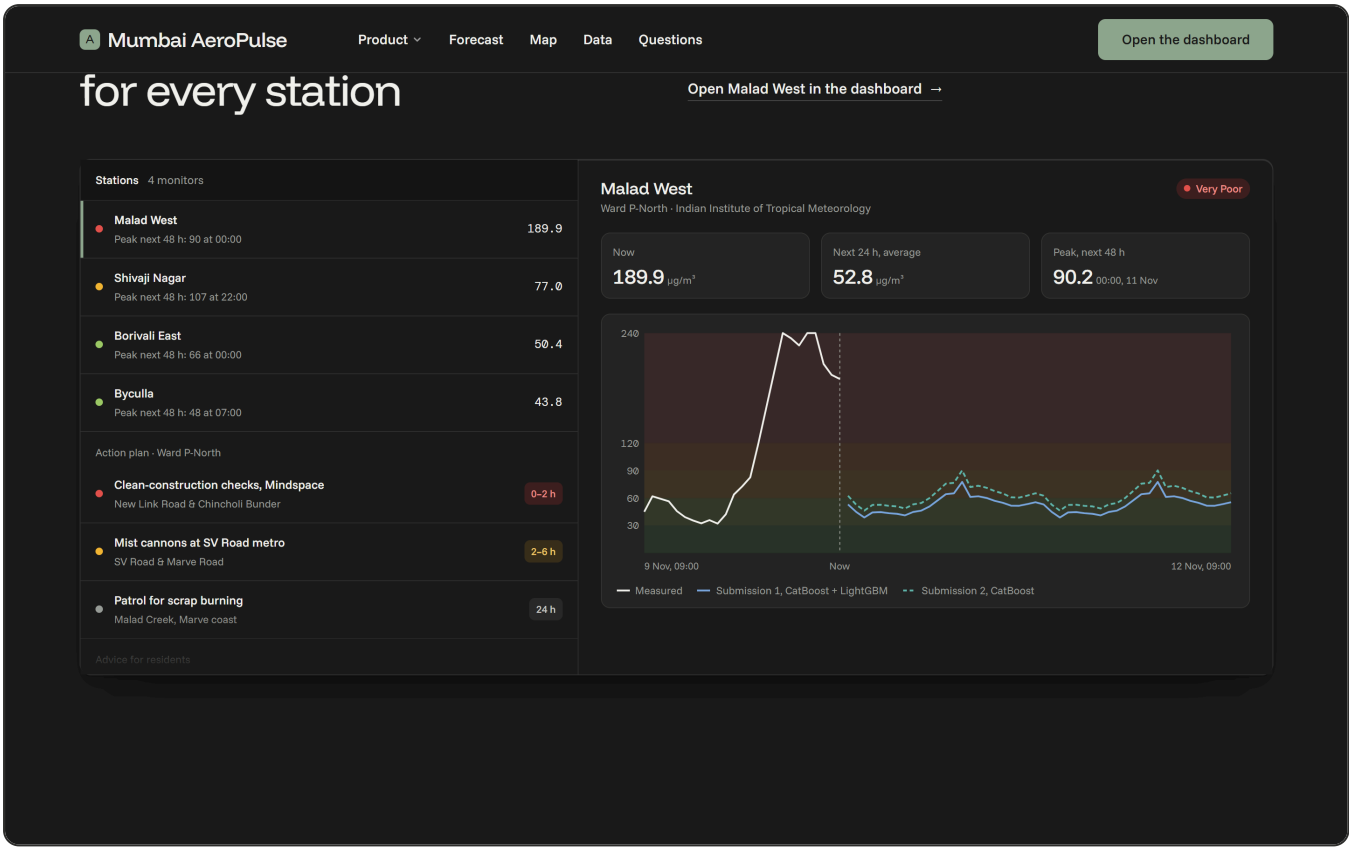


Weather behind the forecast

City-wide hourly weather from Open-Meteo. The ventilation index is boundary-layer height multiplied by wind speed: when it is low, pollution stays near the ground. At 10 Nov, 09:00 the boundary layer was **170 m** deep.

± 3 days | ± 7 days | All 44 days

Landing page: one station's forecast and its ward action plan



How it works

Data

- Hourly PM2.5 from the challenge dataset (CPCB CAAQMS stations run by IITM and the BMC).
- Hourly weather for Mumbai from Open-Meteo (CC BY 4.0).
- `scripts/prepare_dashboard_data.py` builds static JSON for the site: 648 hourly points per station with both forecasts, 615 days of daily history and 1,056 hours of weather.
- Snapshot cutoff: 10 Nov 2026, 09:00 IST. Forecasts run to 21 Nov. All times IST.

Models

- **Submission 1:** 50% CatBoost + 50% LightGBM.
- **Submission 2:** CatBoost only, the more conservative of the two.
- Features include lagged and rolling PM2.5, hour and day cycles, and the boundary-layer weather. `reproduce_submission.py` rebuilds both submission files.
- The dashboard shows the gap between the two models per station, so disagreement is visible rather than hidden.

The advisory

The advisory reads the station's current reading, its next 24 hours from both models, and the boundary-layer weather. With a Gemini API key, a Gemini model writes it against a fixed JSON structure. Without one, a rule-based engine in the site (`syntheticAdvisoryEngine.ts`) writes it from the same inputs and named places in each ward. The site always says which of the two produced the text, and states that it is not advice from a public authority.

Runs without any API keys

The public site needs no keys: the built-in map replaces Google Maps, Google Air Quality values are shown as estimates, and the rule-based engine writes the advisory, each labelled as such. With keys set in `dashboard/.env`, the same site uses Google Maps, the Google Air Quality API and Gemini.

Google tools used

Tool	Used for
Google Maps JavaScript API	The station map, dark styled, with markers coloured by CPCB category.
Google Air Quality API	Modelled PM2.5, other pollutants, Universal AQI, India NAQI and health guidance at each station's coordinates, beside the ground reading.
Gemini API	Writing the municipal action plan and residents' advice from the forecast and weather.

Running it

Open the live website above: the landing page is at the root and the dashboard at `/app/`. To run it locally:

`cd dashboard` `npm install` `npm run dev` then open `http://localhost:5173`.

Built with React 18, Vite 5, TypeScript, Tailwind CSS and Recharts. The repository holds the dashboard, the data-preparation and model scripts, and both submission files.

Limits

- The site shows a fixed snapshot of the data, not a live feed; the snapshot time is shown on every page.
- Four stations cannot describe the whole city; values are for the area around each monitor.
- The action plans and advice are suggestions built from the forecast, not instructions from the BMC or any health authority.

Air-quality data: challenge dataset (CPCB stations run by IITM and the BMC). Weather: Open-Meteo, CC BY 4.0. Google Maps, Gemini and Google Cloud are trademarks of Google LLC.